

## On the performance of a tunable grating-based high sensitivity unidirectional plasmonic sensor: supplement

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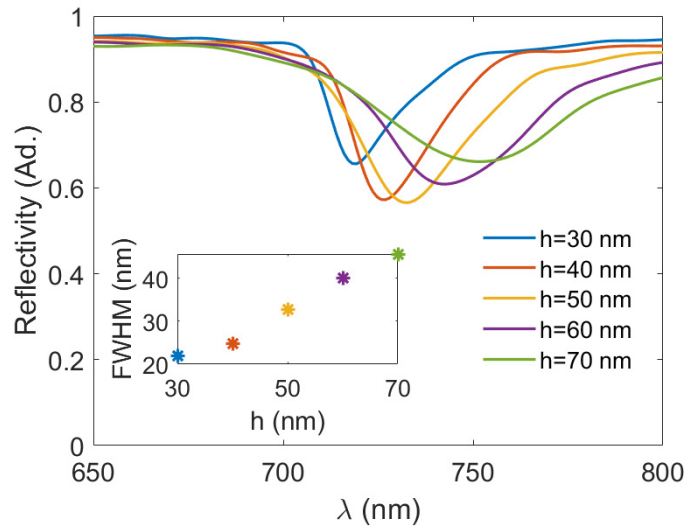
# On the performance of a tunable grating-based high sensitivity unidirectional plasmonic sensor

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One of the key features in the grating fabrication process is the cell height control. To check the impact that this parameter has on the performance of the proposed grating sensor, we have calculated the reflectivity spectrum for a set of unit cell heights  $h \in [30,70]$  nm.

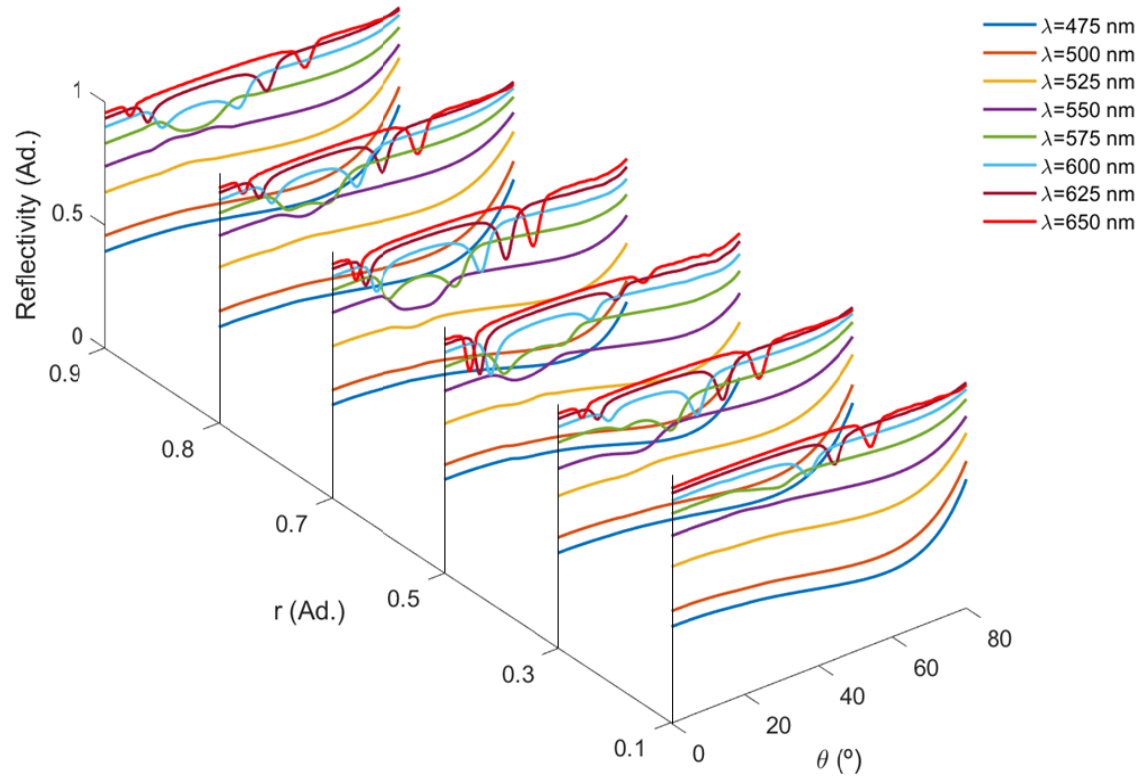


**FIG. S1.** Grating reflectivity versus excitation wavelength for different unit cell heights from 30 to 70 nm. The full width at half maximum is shown in the inset for each case.

As Fig. S1 shows, the reflection drop is redshifted when the cell height grows. This can be explained in terms of the effective refractive index of the grating. When cell height grows, the amount of gold increases with respect to the dielectric, and this results in a growing of the effective refractive index.

Another interesting feature in Fig. S1 is that the Full Width at Half Maximum (FWHM) increases when  $h$  grows showing a maximum increment of around 50 nm. Regarding equation (4), it can be seen how the redshift balances FWHM variations, keeping the Q-factor almost constant.

Figure S2 shows the reflectivity of a complete set of duty cycles with respect to the incidence angle. A subset of the original set ranging from 0.1 to 0.9 in 0.2 steps is shown for simplicity. The  $r = 0.8$  case is also included due to its relevance in the optimization process.



**FIG. S2.** Grating reflectivity versus angle of incidence for a set of wavelengths in the optical range and a set of duty cycles ranging from 0.1 to 0.9. The period is 700 nm. The legend corresponds to the different illumination wavelengths.